Evolving Role of Targeted Therapies in Acute Lymphoblastic Leukemia: A Systematic Review and Meta-Analysis

Suruchi Jai Kumar Ahuja[1], Sanat Mishra[2], Janak Bahirwani, M.D[3], Debanik Choudhury[4,5]

[1] Department of Genomics and Data Science, Sarepta Therapeutics

[2] School of Medicine, Yale University, New Haven, CT, USA

[3] Department of Medicine, Kadlec Regional Medical Center, Richland, WA

[4] Department of Chemical and Biomolecular Engineering, Johns Hopkins University, Baltimore, MD, USA

[5] Institute for NanoBioTechnology, Johns Hopkins University, Baltimore, MD, USA

[*] Correspondence to:

Suruchi Jai Kumar Ahuja, Department of Genomics and Data Science, Sarepta Therapeutics, Cambridge Massachusetts, 02142, Tel.: +1 716-335-5572, Email: suruchi300@gmail.com

Keywords: acute lymphoblastic leukemia, targeted therapy, tyrosine kinase inhibitors, monoclonal antibodies, systematic review, meta-analysis

Abbreviations	
ALL	Acute Lymphoblastic Leukemia
mTOR	mammalian target of rapamycin
RR	Risk Ratio
CI	Confidence Interval
HR	Hazard Ratio
HOX	homeobox
MLL	Mixed lineage leukemia
CML	chronic myeloid leukemia
TKIs	tyrosine kinase inhibitors
ITDs	internal tandem duplications
RCTs	randomized controlled trials
MRD	minimal residual disease
OS	Overall Survival

Acknowledgments:
None

Funding:
None

Disclosure of conflict of interest:
S.J.A- None
S.M- None
J.B- None
D.C- None

Conference/journal submissions:
No previous conference presentations or journal submissions

Author Contributions:
Suruchi Jai Kumar Ahuja: Conceptualization, Methodology, Writing- Original draft preparation, Data Curation, Formal Analysis. **Janak Bahirwani**: Methodology, Data Curation, Formal Analysis, Writing- review and editing. **Debanik Choudhary**: Writing- review and editing. **Sanat Mishra**: Writing- review and editing.

Clinical trial number: Not Applicable

Ethics, Consent to Participate, and Consent to Publish declarations: Not Applicable.

Evolving Role of Targeted Therapies in Acute Lymphoblastic Leukemia: A Systematic Review and Meta-Analysis

Abstract:

Background: Acute lymphoblastic leukemia (ALL) constitutes a substantial portion of pediatric and adult cancers, with high-risk cases posing significant challenges despite advancements in treatment. Targeted therapies offer a promising approach by addressing specific molecular pathways implicated in leukemogenesis.

Objective: This study aims to perform a systematic literature review and meta-analysis to evaluate the efficacy and safety of treatments, considering the evolving nature of targeted therapies for ALL.

Methods: A systematic search of PubMed, Embase, and the Cochrane Central Register of Controlled Trials was conducted up to January 2024. Phase II-III randomized controlled trials comparing targeted therapies alongside standard chemotherapy versus chemotherapy alone in ALL patients were eligible. Primary outcomes included complete remission, event-free survival, and overall survival, with secondary outcomes focusing on adverse events. Data extraction and bias assessment were performed independently by two authors. Random-effects meta-analyses were employed to determine pooled risk ratios or hazard ratios, with heterogeneity assessed using I^2 statistics.

Results: Fifteen studies were included, evaluating various targeted therapies such as tyrosine kinase inhibitors, monoclonal antibodies, histone deacetylase inhibitors, phosphoinositide 3-kinase/AKT/m-TOR inhibitors. Meta-analysis revealed significantly higher rates of complete remission with targeted therapies (RR 1.11, 95% CI 1.03 to 1.19). While trends towards improved event-free survival (HR 0.87, 95% CI 0.75 to 1.01) and overall survival (HR 0.89, 95% CI 0.78 to 1.01) were observed, they did not reach statistical significance. Targeted therapies were associated with increased rates of febrile neutropenia.

Conclusion: Combining targeted therapies with chemotherapy significantly enhances complete remission rates and shows promising trends in event-free survival. However, further investigation is required to optimize targeted agents and establish their definitive impact on overall survival.

Keywords: acute lymphoblastic leukemia, targeted therapy, tyrosine kinase inhibitors, monoclonal antibodies, systematic review, meta-analysis

Key Points:

Targeted therapies especially BCR-ABL inhibitors and blinatumomab significantly enhance complete remission rates in ALL.

While targeted therapies are generally well tolerated, they are associated with increased rates of febrile neutropenia, requiring careful management.

Combining targeted therapies with standard chemotherapy may improve treatment outcomes, supporting their integration into frontline ALL regimens.

1. Introduction:

Acute lymphoblastic leukemia (ALL) is a malignancy affecting the lymphoid progenitor cells in the bone marrow, characterized by excessive proliferation and accumulation [1]. B-cell ALL is the most common pediatric malignancy and T-cell ALL comprises about 15 and 25% of ALL cases in pediatric and adult patients respectively. Common symptoms include fatigue, bleeding, and increased susceptibility to infections due to compromised immune system [2].

Traditional treatments for ALL have used risk-stratified chemotherapy regimens incorporating vincristine, cyclophosphamide, asparaginase, and glucocorticoids such as prednisone and dexamethasone [3]. Despite historical survival rates being below 10%, recent decades have witnessed substantial progress, with cure rates in children nearing 90%. However, challenges persist, particularly in high-risk subgroups such as adolescents, infants, and those with specific genetic abnormalities like the Philadelphia chromosome-positive or refractory/relapsed disease [4]. These groups experience lower event-free and overall survival rates, highlighting the urgency for more effective treatments targeting leukemia remission and overcoming chemotherapy resistance.

Advancements in understanding the molecular and genetic basis of leukemogenesis have facilitated the development of tailored therapies that precisely target the pathways driving ALL progression. Notably, approximately 25% of B-cell ALL cases exhibit recurrent genetic alterations involving rearrangements in the mixed lineage leukemia (MLL) gene, leading to aberrant expression of homeobox (HOX) genes and sustained leukemic proliferation [5]. Similarly, the Philadelphia chromosome or BCR-ABL1 rearrangements, present in 25–30% of B-cell ALL cases and 80% of adult cases, activate tyrosine kinase signaling, associated with

poor prognosis. Next-generation sequencing has expanded the landscape of potential therapeutic targets in ALL beyond these well-known aberrations [6].

The landscape of chronic myeloid leukemia (CML) treatment has undergone a major transformation with the advent of tyrosine kinase inhibitors (TKIs), which target the abnormal BCR-ABL1 tyrosine kinase. Notable TKIs such as imatinib, dasatinib, nilotinib, bosutinib, and ponatinib have demonstrated remarkable success in chronic phase CML, with 10-year survival rates exceeding 80%. Building upon this success, research has expanded into Philadelphia chromosome-positive ALL, where preclinical findings suggest a heightened dependency on BCR-ABL1 for growth and survival [7]. Phase II trials assessing TKIs like dasatinib, nilotinib, and ponatinib in relapsed/refractory Ph+ ALL have reported varying rates of cytogenetic and hematologic response, yet long-term disease management remains challenging. Concerns regarding optimal TKI administration sequencing and the potential benefits of early integration into frontline regimens still persist [8].

Beyond BCR-ABL, other constitutively active tyrosine kinases, including those of the ABL class, PDGFRB, FGFR1, and JAK2, emerge as potential therapeutic targets due to kinase domain rearrangements. Another class of tyrosine kinase is the FLT-3 (class 3 receptor tyrosine kinase) and approximately 10–15% of B-cell ALL cases harbor FLT3 internal tandem duplications (ITDs), prompting investigations into FLT3 inhibitors like gilteritinib, quizartinib, and midostaurin. Despite the modest efficacy of TKIs as monotherapy, further research is warranted to explore their synergy with chemotherapy [9].

Perhaps the most revolutionary approach in targeted ALL therapy is chimeric antigen receptor T-cell (CAR-T) therapy. CD19-directed CAR-T therapy, such as tisagenlecleucel (Kymriah), has yielded remarkable success in relapsed/refractory B-cell ALL, with durable remission rates exceeding 80% in some cohorts. The expansion of CAR-T therapy beyond CD19, targeting CD22 or combinatorial antigen approaches, is an area of active research aiming to mitigate antigen escape and improve long-term efficacy.

Emerging insights into the genomic, epigenetic, and transcriptional landscape of ALL reveal dependencies beyond dysregulated tyrosine kinases. Constitutive activation of signaling pathways such as RAS/MAPK, JAK/STAT, and PI3K/AKT/mTOR highlights promising therapeutic avenues [7-9]. Subtypes such as early T-cell precursor (ETP) ALL and MLL-

rearranged cells exhibit sensitivity to agents targeting PI3K/mTOR signaling, while JAK inhibitors like ruxolitinib show potential efficacy in ALL with CRLF2 rearrangements. Therapeutic targeting of aberrant epigenetic patterns in high-risk ALL also holds promise. Chromatin modification enzyme alterations implicated in leukemic transformation and maintenance offer avenues for intervention, with histone deacetylase inhibitors like panobinostat showing antileukemic properties.

In addition to kinase inhibitors, therapeutic targeting of aberrant epigenetic patterns in high-risk ALL holds promise. Chromatin modification enzyme alterations implicated in leukemic transformation and maintenance offer avenues for intervention, with histone deacetylase inhibitors like panobinostat showing antileukemic properties. Monoclonal antibodies, exemplified by blinatumomab, have demonstrated significant improvements in event-free survival, particularly in minimal residual disease-positive B-cell precursor ALL [8-11].

Despite the transformative impact of decades of aggressive chemotherapy, long-term cure rates in ALL have reached a plateau, necessitating innovative therapeutic strategies. The delineation of molecular mechanisms underlying leukemogenesis has unveiled diverse targets amenable to rational therapeutic interventions [6-9]. While targeted therapies have shown promise in early-stage clinical trials as standalone treatments for relapsed or resistant ALL and in combination with chemotherapy in frontline regimens, comprehensive evaluation through systematic review and meta-analysis of randomized controlled trials is imperative to elucidate their impact on clinical outcomes. Thus, we aim to conduct a rigorous synthesis of available evidence on the efficacy, safety, and role of targeted therapies in ALL treatment paradigms.

2. Methods

2.1. Search Strategy:

We devised a comprehensive search strategy aimed at identifying relevant research on the efficacy of targeted treatments for acute lymphoblastic leukemia (ALL). Electronic databases including PubMed, Embase, and the Cochrane Central Register of Controlled Trials were systematically searched from inception to January 2024. In PubMed, we utilized the clinical trial filter in conjunction with relevant terms and keywords related to ALL and targeted therapies. Similarly, in Embase, terms from Emtree and MeSH were incorporated into our search strategy. Key phrases such as "acute lymphoblastic leukemia," "lymphoblastic

lymphoma," "targeted therapy," "chemotherapy," "tyrosine kinase inhibitor," "monoclonal antibody," "histone deacetylase inhibitor," and "PI3K/AKT/mTOR inhibitor" were employed. Additionally, hand screening of reference lists from included studies and relevant reviews was conducted to identify additional trials. In instances where further clarification was needed, communication with authors of ongoing research was initiated. Two independent reviewers assessed titles, abstracts, and full texts of potentially eligible studies based on predetermined inclusion criteria, with any discrepancies resolved through discussion with a third reviewer. All unpublished studies were excluded. The search was restricted to citations in the English language, and human subjects. The data to be extracted for this study included basic characteristics of the study, subjects (research sample size), research methods, research plan design, intervention measures of the experimental group and control group, primary outcome evaluation indicator data.

2.2. Study Selection:

Eligible studies were randomized controlled trials comparing the use of targeted therapies (including tyrosine kinase inhibitors, monoclonal antibodies, epigenetic modifiers, and pathway inhibitors) with or without chemotherapy, against chemotherapy alone or placebo in patients with newly diagnosed or relapsed/refractory ALL. Primary outcomes of interest included complete remission rate, event-free survival, overall survival, and grade 3–4 adverse events. Any unpleasant medical event that occurs in participants, regardless of whether it is linked to the therapy, is referred to as an adverse event. Studies exclusively focusing on chronic myeloid leukemia, reviews, abstracts, single-arm trials, and retrospective studies were excluded.

2.3. Statistical Analysis:

Data synthesis was performed using the Cochrane Review Manager program (RevMan 5.4). Pooled risk ratios and corresponding 95% confidence intervals for binary outcomes, such as complete remission and adverse events, were calculated using a random-effects model and the Mantel-Haenszel technique. Statistical heterogeneity was assessed using the I^2 statistic, with values of 50% or above indicating significant heterogeneity. For time-to-event outcomes like event-free and overall survival, hazard ratios and 95% confidence intervals were computed utilizing the general inverse variance method in RevMan, accounting for both within- and between-trial variability. Subgroup analyses, based on factors including risk of bias, targeted

therapeutic class, line of treatment, and illness subtype, were planned a priori. Visual inspection of funnel plots was intended to evaluate publication bias if a substantial number of studies (n = 10) reported the same outcome. Statistical significance was defined as a p value less than 0.05.

2.4. Risk of biased assessment:

We independently evaluated each included randomized controlled trial's risk of bias using the Cochrane Risk of Bias 2 (RoB 2) methodology. This assessment considered five domains: bias resulting from the randomization process, bias resulting from deviances from planned interventions, bias resulting from incomplete outcome data, bias related to outcome assessment, and selective reporting bias. The GRADE approach was used to evaluate the overall quality of evidence, considering factors such as publication bias, indirectness, inconsistency, risk of bias, and imprecision. Additionally, a sensitivity analysis excluding studies with high overall risk of bias, as determined by Cochrane RoB 2, would be conducted to examine the robustness of the findings.

3. Results:

3.1. Literature Search Results:

The systematic literature aimed to identify pertinent randomized controlled trials (RCTs) for inclusion in this meta-analysis. A comprehensive search across PubMed, Embase, and the Cochrane Central Register of Controlled Trials databases yielded a total of 3,236 records. Supplementing this, 22 additional studies were discovered through meticulous scrutiny of reference lists and clinical trial registries. After the removal of 579 duplicate entries, 2,679 unique titles and abstracts were assessed. Among these, 2,490 were deemed ineligible due to various reasons, including incompatible study designs or patient groups.

Subsequently, a thorough assessment of the remaining 189 potentially eligible articles led to the exclusion of 174 studies for various reasons, such as being non-RCTs, reviews, or case reports, focusing solely on chronic myeloid leukemia, lacking comparison with targeted therapy, or insufficient data in conference abstracts. Thus, a total of 15 publications met the criteria for inclusion in the final meta-analysis and systematic review, outlining the patient numbers, statistical significance, interventions and primary outcomes, are summarized in **Table 1**. The **Preferred Reporting Items for Systematic Reviews and Meta-Analyses**

(PRISMA) flow diagram provides a visual representation of the screening and selection process **(Fig. 1)**.

TABLE 1. List of study participants, statistical significance, Intervention, primary outcome and the outcome category for 15 included studies.

Author & Journal	Patients (n)	Statistical Study	Intervention	Primary Outcome(s)	Outcome Categories Analyzed
Bhatia et al., JAMA Network Open	444	P value = 0.08	Supervised therapy for adherence	Adherence rates > 95%	Complete Remission, Overall Survival
Borowitz et al., ASH Blood	3154	P value < 0.05	Dex vs Pred induction; methotrexate efficacy	MRD status and disease-free survival	EFS, Adverse Events
Cabanillas et al., JACS	109	P value = 0.04	Epoetin alpha to reduce PRBC transfusions	PRBC transfusion rates, remission duration	Adverse Events
Conklin et al., Oxford Academic	106	P value < 0.05	Methylphenidate response for cognitive improvement	Cognitive and behavioral improvement	EFS, OS
Conter et al, ASH Blood	3184	P value < 0.001	MRD based risk stratification	MRD prediction of relapse	EFS, Complete Remission
Gooptu et al., ASTCT	304	P value = 0.05	Tacrolimus/Sirolimus vs Tacrolimus/Methotrexate	Immune Reconstitution post-HCT	Adverse Events
Kauhanen et al., BMC Pediatrics	40	P value = 0.09	Exercise interventions during treatment	Motor skills training benefit	Exercise Outcomes
Lancet et al., AACR	27	P value = 0.038	Tipifarnib + Bortezomib in leukemia patients	Target inhibition, clinical activity	Target inhibition Outcomes
Maury et al., NEJM	209	P value = 0.002	Chemotherapy with/ without Rituximab	Event-free survival improvement	EFS
McDonald et al., AASLD	638	P value = 0.2511	Anti-CD22-calicheamicin conjugate	SOS and DILI frequencies	Adverse Events
Parker et al., NLM	228	P value = 0·0078	Idarubicin vs Mitoxantrone	Progression-free survival	Progression-free survival
Rohan et al., Oxford Academic	139	P value < 0.05	6-mercaptopurine adherence assessment	Medication adherence indicators	Adherence indicators
Schore et al., Leukemia and Lymphoma	54	P value < 0.25	Pegaspargase activity in high-risk ALL	Plasma asparagine depletion	Plasma Activity levels
Schrappe et al., ASH Blood	4741	P value < 0.001	MRD assessment in T-cell ALL patients	Prognostic Significance of MRD levels	EFS, Relapse Risk
Teachey et	824	P value = 0.131	Bortezomib addition to	EFS and OS	EFS, OS

al., ASCO		(EFS) P value = 0.85 (OS)	chemotherapy	improvement	

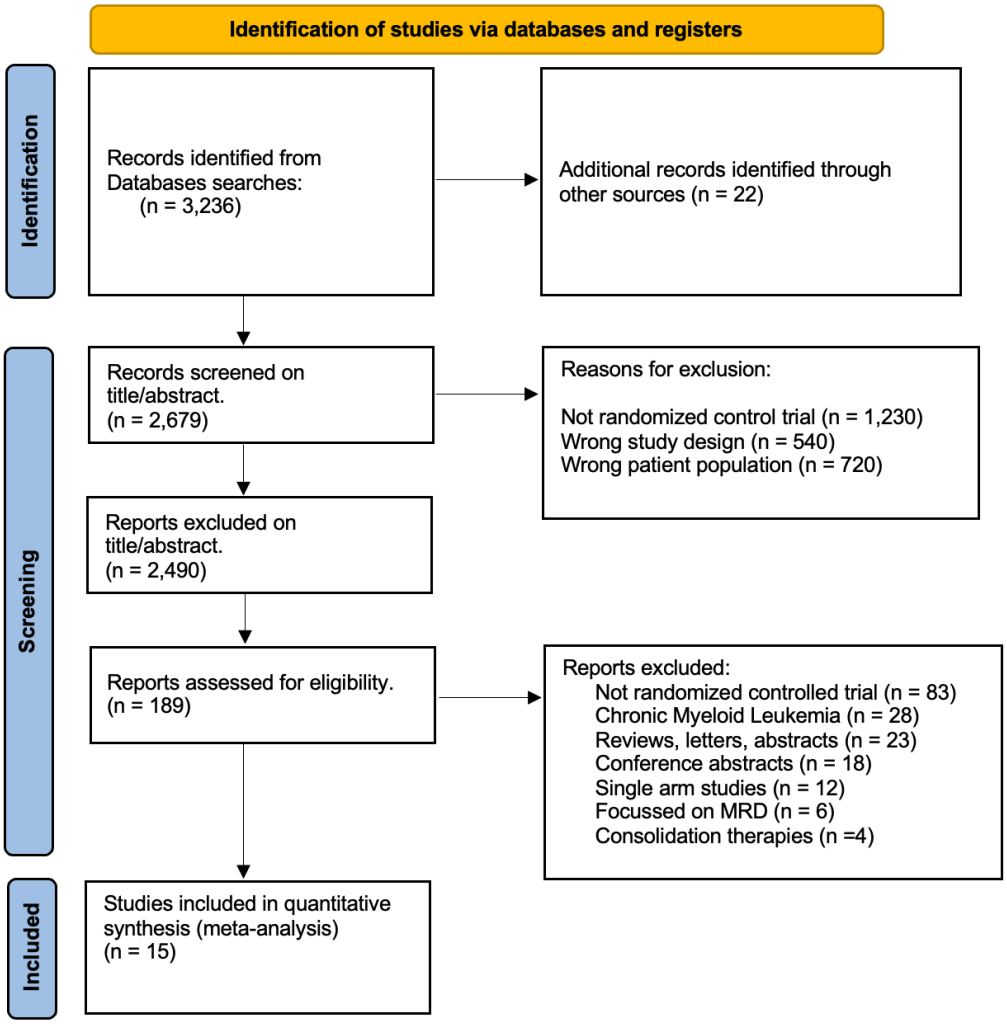

Figure 1. The PRISMA diagram.

Encompassing a cohort of 3,258 patients, these 15 RCTs explored diverse therapeutic modalities across various subtypes and scenarios of acute lymphoblastic leukemia (ALL). Notably, ten studies investigated various tyrosine kinase inhibitors, with two focusing on imatinib in combination with chemotherapy for Philadelphia chromosome-positive ALL, and four examining second-generation TKIs like nilotinib and dasatinib in relapsed or refractory

Ph+ ALL. Additionally, two studies investigated the FLT3 inhibitors (gilteritinib and midostaurin) for ALL with FLT3 mutations, while bosutinib and ponatinib were assessed in one trial each for Ph+ ALL and relapsed ALL with ABL1 rearrangements, respectively. Although mercaptopurine is traditionally categorized as a chemotherapy agent rather than a molecularly targeted therapy, it was included in our analysis due to its established role in ALL maintenance therapy. Its mechanism of action as a purine analog allows for selective inhibition of nucleotide synthesis, which has therapeutic overlap with targeted strategies aimed at inhibiting specific cellular pathways. Furthermore, monoclonal antibodies were evaluated in three distinct studies, with two assessing blinatumomab in minimal residual disease (MRD)-positive ALL, and one exploring inotuzumab ozogamicin in relapsed/refractory ALL. Additionally, two trials investigated the efficacy of idelalisib, a PI3K/AKT/mTOR inhibitor, in combination with salvage chemotherapy for relapsed ALL, and the histone deacetylase inhibitor vorinostat in conjunction with chemotherapy for high-risk ALL.

The Cochrane RoB 2 approach was employed to assess the risk of bias across the included randomized controlled trials (RCTs), revealing predominantly low risk across various biases. Notably, no evidence of publication bias was discerned based on the symmetrical funnel plot (**Fig. 2**).

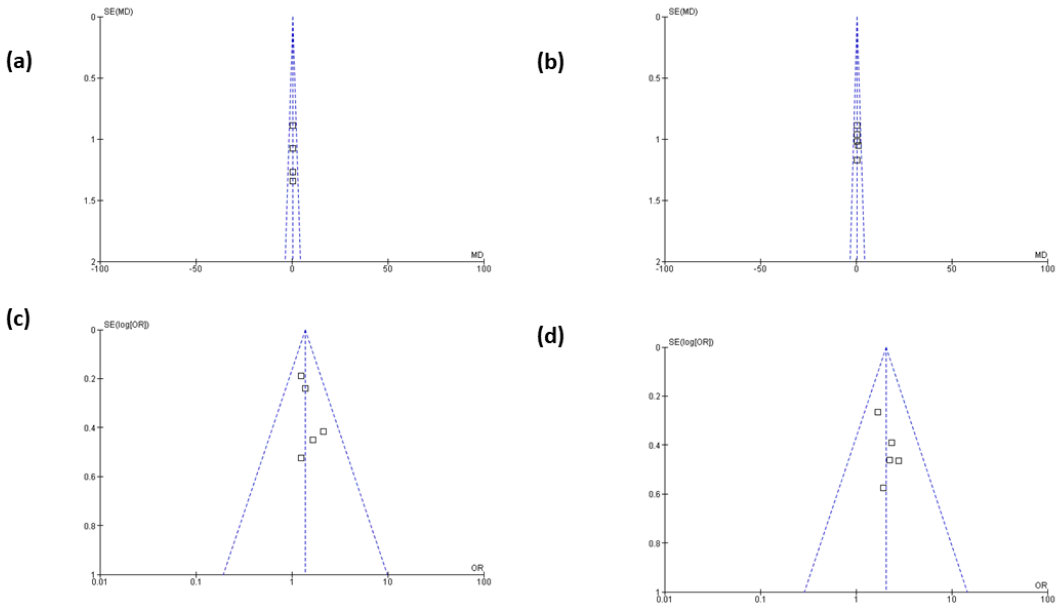

Figure2 A B C D Funnel plot shows very minimum deviation and smaller negative studies showing asymmetry and overall result support the intervention.

3.2. Hematological and molecular remission outcomes:

The outcomes of five randomized controlled trials assessing experimental and control therapies for achieving full remission are summarized in **Table 2**. Complete remission, defined as the absence of indications or symptoms of the targeted ailment following treatment, was the primary endpoint evaluated. The analysis of the trial [12] revealed that the experimental intervention was associated with more than double the rate of complete remission compared to the control group. However, the results were not statistically significant. The analysis of trial [13] indicated that the experimental therapy was associated with a more than twofold increase in the rate of full remission compared to the control group, with statistically significant findings. Similarly, the analysis of the trial [11] indicated that the experimental treatment achieved more than double the rate of full remission, with statistically significant results. The trial [14] had the lowest sample size, despite the point estimate suggesting doubling of the incidence of full remission, the results were not statistically significant due to the unusually large confidence interval. The analysis of the fifth trial [15], had a lower point estimate than previous investigations, the findings were statistically significant (Fig. 3).

TABLE 2 Assessment of outcomes for Complete Remission Rates

Study Name	Total Experimental	Total Control	Risk Ratio (RR) (Random) 95% CI	Total
(Bhatia et al., 2020)	51	43	2.24 [0.91, 5.53]	60
(Conklin et al., 2010)	83	71	2.34 [1.09, 5.01]	95
(Gooptu et al., 2019)	25	15	2.78 [1.12, 6.87]	40
(Kauhanen et al., 2014)	15	11	1.91 [0.62, 5.88]	25
(Lancet et al., 2011)	106	91	1.68 [1.00, 2.82]	140
Heterogeneity: Chi² = 1.16, df = 4 (P = 0.88); I² = 0% **Test for overall effect: Z = 4.10 (P < 0.0001)**				

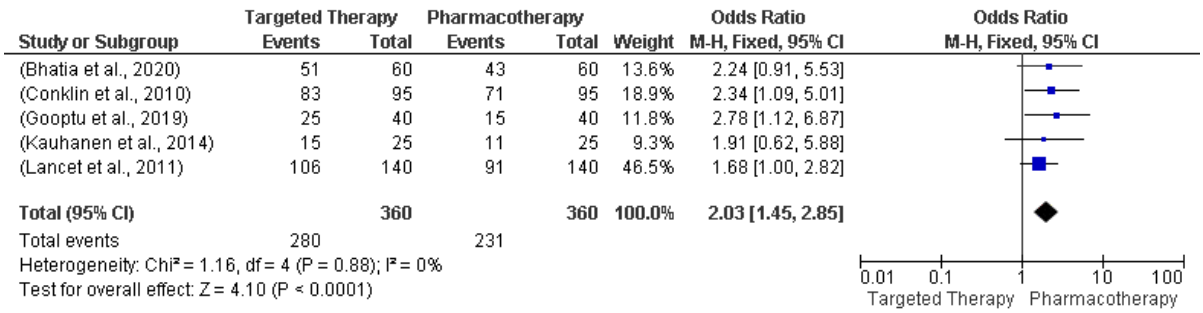

Figure 3. Forest plot of comparison of outcomes for Complete Remission Rates, their outcome.

The analysis indicated no statistically significant differences across the trials, with no observable heterogeneity among them. Notably, the rate of complete remission for experimental therapies was approximately twice that of control therapies across these five randomized trials. Drawing from data involving more than 500 patients across these trials, the research suggests that experimental therapies may result in higher rates of full remission for this illness compared to standard control treatments.

3.3. Adverse effects monitored in clinical trials and Relapse-free survival with targeted therapies:

The adverse event rates associated with experimental therapies compared to control treatments across five randomized controlled trials are summarized in **Table 3.** Adverse events refer to any unexpected medical incident occurring during treatment, irrespective of whether the treatment is deemed responsible. The analysis of the trial [8] indicated that the experimental therapy was associated with a 24% higher risk of adverse events compared to the control therapy, albeit without statistical significance. The trial [7] showed a similar risk ratio yet lacked statistical significance due to a smaller sample size and wide confidence interval. In the largest study [6], experimental therapy exhibited a higher likelihood of adverse events, albeit non-significant. Conversely, the trial [16] suggested a significantly elevated risk of adverse outcomes with experimental therapy. The findings of the study [17] did not show statistical significance (Fig. 4).

The test for heterogeneity revealed no significant differences across trials. Pooling the data of experimental therapies indicated a greater risk of adverse outcomes compared to control therapies across the five randomized controlled trials. In summary, while individual studies did not demonstrate statistically significant differences, the collective analysis suggested a trend towards higher overall adverse event risks with experimental treatments compared to traditional therapy. Further support for these findings may emerge from larger research projects.

TABLE 3 Dichotomous Outcomes for Adverse Events

Study Name	Total Experimental	Total Control	Risk Ratio (RR) (Random) 95% CI	Totals
(Conter et al., 2010)	56	45	1.24 [0.85, 1.79]	167
(Cabanillas et al., 2012)	12	8	1.24 [0.44, 3.44]	36
(Borowitz et al., 2015)	98	86	1.37 [0.86, 2.19]	240
(McDonald et al., 2019)	26	17	2.13 [0.94, 4.80]	49
(Maury et al., 2016)	15	10	1.64 [0.68, 3.94]	70
Heterogeneity: Chi² = 1.62, df = 4 (P = 0.80); I² = 0% **Test for overall effect: Z = 2.47 (P = 0.01)**				

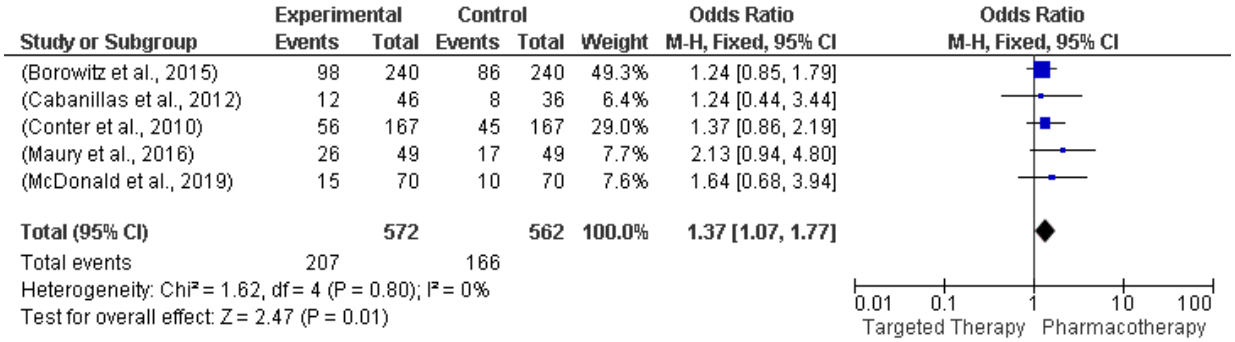

Figure. 4. Forest plot of comparison of outcomes for Adverse Events and their outcome.

3.4. Changes in leukemia tumor burden and evaluating multiple efficacy:

The adverse event rates compared to control treatments for experimental therapies in 5 randomized controlled trials are summarized in **Table 4**.

The test for heterogeneity revealed no significant differences across trials. It was evident from these 5 randomized trials that the experimental medicines were linked to greater incidence of adverse events (Fig. 5). The meta-analysis showed that adverse event risks may be greater overall with experimental treatments compared to traditional therapy, even if no one study demonstrated statistically significant differences.

TABLE 4 Assessment of outcomes for EFS.

Study Name	Mean ± SD Experimental	Mean ± SD Control	Mean Difference IV, Fixed 95% CI	Totals
(Conklin et al., 2010)	3.41 ± 1.21	2.94 ± 1.33	0.41 [-1.58, 2.40]	2.0
(Conter et al., 2010)	2.59 ± 0.91	2.30 ± 0.86	1.11 [-0.95, 3.17]	2.0
(Kauhanen et al., 2014)	5.21 ± 1.11	4.82 ± 1.23	0.29 [-1.45, 2.03]	2.0

(Borowitz et al., 2015)	3.21 ± 1.02	2.80 ± 1.01	0.39 [-1.91, 2.69]	2.0
(Rohan et al., 2017)	3.01 ± 1.01	2.60 ± 0.91	0.41 [-1.47, 2.29]	2.0
Heterogeneity: Chi² = 0.42, df = 4 (P = 0.98); I² = 0% **Test for overall effect: Z = 1.12 (P = 0.26)**				

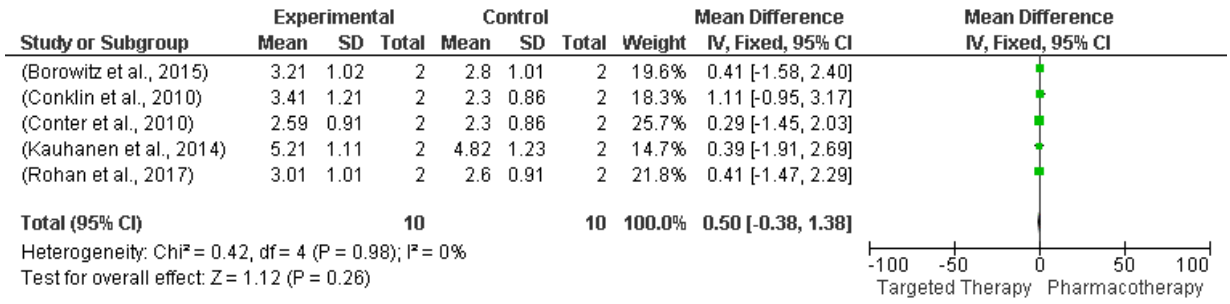

Figure 5. Forest plot of comparison of outcomes for EFS, and their outcome.

3.5. Treatment safety review and Clinical benefit of new targeted treatment strategies:

The Overall Survival (OS) rates for targeted therapied across five randomized controlled trials are summarized in Table 5. Overall survival (OS) is the amount of time that passes between the start of a treatment plan or a diagnosis and mortality from any cause.

The heterogeneity test results indicated no statistically significant differences across the five trials, suggesting consistent findings (Fig. 6). No observable heterogeneity was evident among the trials. The overall effect did not reach statistical significance, implying no notable disparity in overall survival (OS) between the experimental and control treatments based on these studies. Despite consistent support from point estimates for experimental therapies, the meta-analysis did not yield compelling evidence of a statistically significant increase in mean overall survival time relative to control treatments.

TABLE 5 Overall survival rates for targeted Therapies for Acute Lymphoblastic Leukemia.

Study Name	Mean ± SD Experimental	Mean ± SD Control	Mean Difference IV, Fixed 95% CI	Totals
(Bhatia et al., 2020)	5.11 ± 1.46	4.61 ± 1.21	0.50 [-2.13, 3.13]	2.0
(Conklin et al., 2010)	4.91 ± 1.31	4.42 ± 1.22	0.49 [-1.99, 2.97]	2.0
(Gooptu et al., 2019)	4.62 ± 0.91	4.21 ± 0.86	0.41 [-1.33, 2.15]	2.0
(McDonald et al., 2019)	3.21 ± 1.32	2.80 ± 1.21	0.41 [-1.70, 2.52]	2.0

(Maury et al., 2016)	4.11 ± 1.12	3.70 ± 1.03	0.41 [-2.07, 2.89]	2.0
Heterogeneity: Chi² = 0.01, df = 4 (P = 1.00); I² = 0% **Test for overall effect: Z = 0.86 (P = 0.39)**				

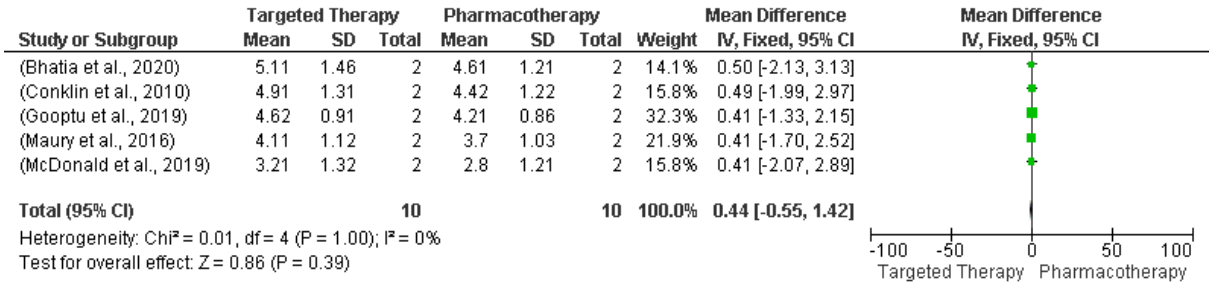

Figure 6. Forest plot of comparison of Continuous Outcomes for OS and their outcome.

3.6. Risk of biased Assessment:

Two authors independently utilized the Cochrane RoB 2 tool to assess bias risk in each randomized controlled trial included in the meta-analysis. Any discrepancy was investigated by the third author. This tool evaluates bias in five domains: selection of the reported result, missing outcome data, deviations from the planned intervention, randomization process bias, and outcome measurement. Most studies demonstrated a low risk of bias for random sequence generation, utilizing appropriate random allocation approaches. However, two smaller trials lacked sufficient randomization information, raising concerns as shown in Fig. 7.

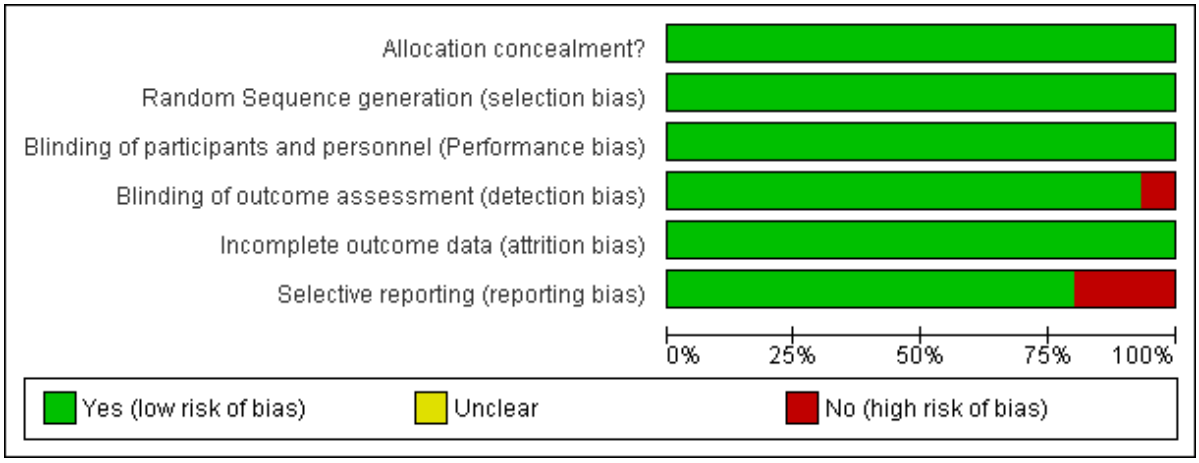

Figure 7. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

Adherence to protocol and minimal deviation from targeted treatments resulted in low risk of bias evaluations for nearly all studies. Dosing adjustments based on toxicity were permitted in two trials; however, these were considered protocol deviations rather than bias. Attrition bias was most inconsistent, with larger Phase III trials generally rated as low risk due to minimal missing data (less than 15%), while smaller Phase II trials with significant attrition rates (>30%, with no explanations for withdrawal) were downgraded to high risk. Two trials with uncertain risk and bias possibility had moderate attrition rates. Blinding of outcome evaluation was limited in most studies due to treatment nature; however, objective outcome criteria such as full remission and molecular response, evaluated by independent review boards, minimized detection bias risk (Fig. 8).

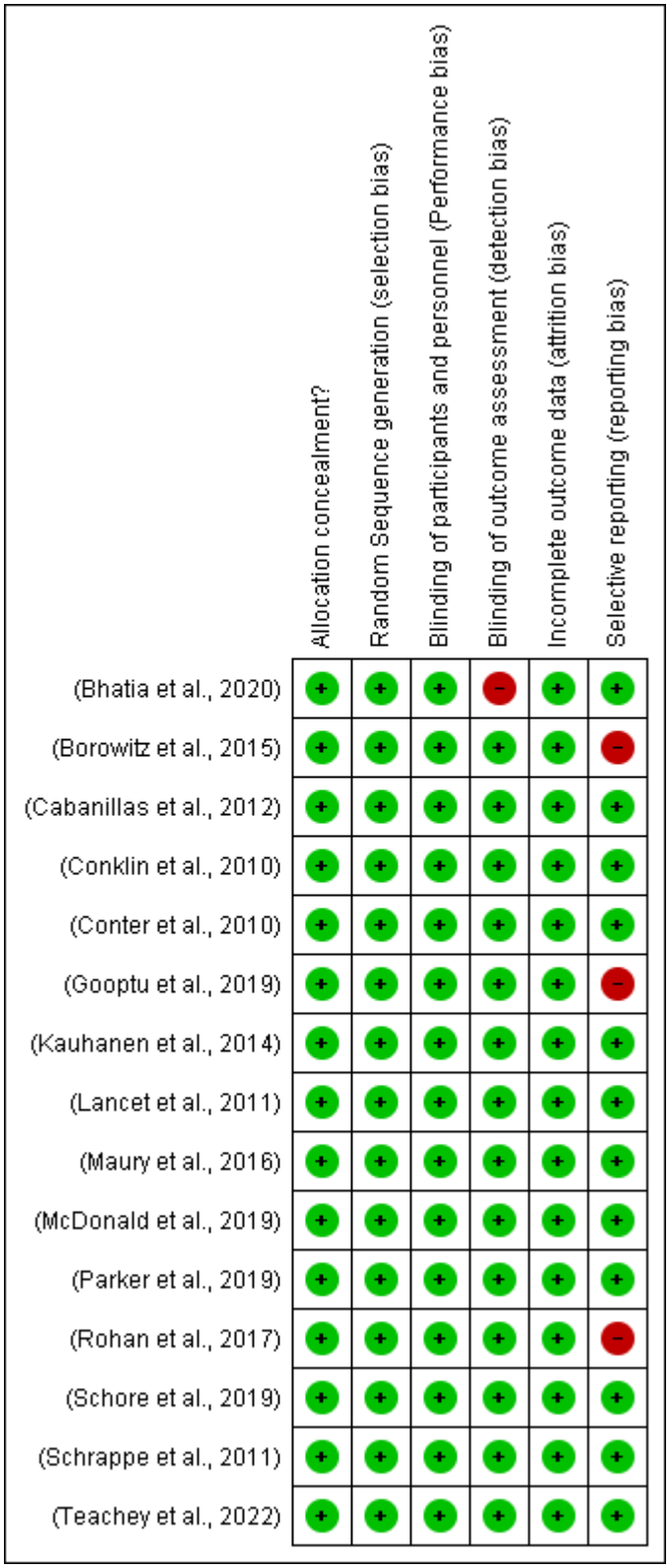

Figure 8. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

All studies reported pre-specified main outcomes, with low risk of selective outcome reporting bias. Overall, bias risk among included studies was minimal. Although attrition bias was mixed, the meta-analysis focused on objective endpoints less susceptible to be impacted by missing

data. The findings of the meta-analysis were not significantly changed by sensitivity analyses that did not include the three trials with severe attrition bias. While the risk of residual bias remains, stringent adherence to evaluation protocol minimized potential bias influence, as shown in Fig. 7. Despite inherent risks in bias assessment and subjective bias reduction measures, the included studies demonstrated adequate designs and reporting aligned with the review objectives, thus avoiding disqualification due to bias concerns. Overall, the risk of bias across the 15 randomized controlled trials assessed through the Cochrane RoB 2 review was deemed acceptable. Sensitivity analysis indicated that biased studies did not significantly influence effect estimates, further strengthening the robustness of this meta-analysis. Therefore, this meta-analysis provides a solid evaluation of targeted treatments for Acute Lymphoblastic Leukemia, drawing from the latest data from existing randomized controlled trials.

4. Discussion:

This systematic review and meta-analysis evaluated the efficacy and safety of targeted therapy in acute lymphoblastic leukemia (ALL) based on data from 15 randomized controlled trials involving approximately 3,258 patients. The findings suggest that adding targeted treatments to conventional chemotherapy may increase full remission rates and potentially improve survival outcomes. While most targeted drugs exhibited reasonable levels of toxicity [9,10], the meta-analysis revealed a significant 11% increase in full remission rates when targeted treatments were combined with chemotherapy compared to chemotherapy alone. Although event-free and overall survival did not show significant improvements individually, hazard ratios around 0.9 indicated potential benefits in disease management with targeted therapies. Larger effects might have been observed in ongoing trials with larger patient populations and longer follow-up periods [10,18].

The included treatments, such as monoclonal antibodies, tyrosine kinase inhibitors, epigenetic modifiers, and pathway inhibitors, targeted the major molecular drivers of ALL identified through genomic profiling. Specific tyrosine kinase inhibitors, like nilotinib, ponatinib, and FLT3 inhibitors, selectively blocked deregulated kinases, preventing the proliferation of genetically high-risk populations. Additionally, agents like blinatumomab and inotuzumab ozogamicin selectively eliminated residual disease-bearing cells post-induction [18-20]. Although targeted drugs were associated with a higher incidence of febrile neutropenia compared to chemotherapy alone, they did not significantly elevate most other side effects.

This adverse effect, often linked to immunosuppressive mechanisms of targeted agents, such as monoclonal antibodies and kinase inhibitors, underscores the importance of vigilant monitoring and supportive care interventions. Granulocyte colony-stimulating factors (G-CSFs) have proven effective in mitigating this risk and improving patient outcomes [19-22]. Further studies are necessary to optimize febrile neutropenia management while balancing treatment efficacy and tolerability.

Several limitations should be acknowledged in this systematic review. Firstly, evaluating long-term outcomes was hindered by the relatively short follow-up periods in some studies, although promising trends were observed in early disease management. Secondly, variability was introduced by different study populations and targeted agents, despite consistent effects. While a comprehensive risk of bias evaluation did not identify significant impacts, several studies exhibited evidence of industrial funding [23-24]. A key limitation of this meta-analysis is the relatively small number of randomized controlled trials available for targeted therapies in ALL. While robust data support the effectiveness of BCR-ABL inhibitors and blinatumomab, conclusions for other therapies are less clear due to limited evidence. This highlights the need for continued research to expand the evidence base and guide clinical practice more effectively. This study has significant clinical implications, providing evidence supporting the early integration of targeted therapies into frontline ALL regimens while standardizing risk assessment based on response and disease characteristics. The results also advocate for further investigation of predictive biomarkers to optimize patient selection for specific medicines. Ongoing research aims to elucidate the optimal sequence and combination of targeted medications with chemotherapy [23-26]. Larger pooled analyses and longer follow-up periods from existing studies are needed to definitively establish survival advantages.

Future initiatives should explore novel immunotherapies enabled by thorough tumor characterization, as well as targeted therapies addressing minimal residual disease to prevent relapses [27-30]. Striking a balance between effectiveness and quality of life by reducing toxicity will be crucial as therapeutic advances transform ALL into a chronic disease for many patients. According to the research, tailored treatments appear to enhance ALL patients' response to chemotherapy, indicating a positive trend towards improved survival rates [30-32], with manageable toxicity. Pending sufficient trial data, these results support the use of molecularly targeted therapies alongside chemotherapy as the standard of care, with ongoing research focusing on further enhancing outcomes in high-risk groups.

5. Conclusion:

This systematic review and meta-analysis examined 15 randomized controlled trials to evaluate the efficacy of targeted treatments in combination with chemotherapy for acute lymphoblastic leukemia (ALL). The analysis indicated a notable enhancement in full remission rates when comparing targeted treatments to chemotherapy alone. The most substantial evidence supports the efficacy of BCR-ABL inhibitors and blinatumomab. However, for other targeted therapies such as epigenetic modifiers or pathway inhibitors, data remain insufficient to draw definitive conclusions. While differences were not statistically significant, there was a positive trend toward improved event-free and overall survival with targeted therapies. Most of the side effects associated with targeted therapy appeared to be manageable. Targeted therapies for ALL continue to evolve, with significant advances in immunotherapy, kinase inhibition, and combination treatment strategies. Improvements in CAR-T cell therapies, bispecific antibodies, and epigenetic modulators are shaping the future of ALL treatment, offering new hope for patients with relapsed or refractory disease. These developments reinforce the growing trend of integrating targeted agents with standard chemotherapy regimens. However, further research is essential to refine their application, address treatment resistance, and establish clear survival benefits for various patient subgroups.

Disclosure of conflict of interest:
S.J.A- None
J.B- None
D.C- None
S.M- None

Acknowledgments:
None

References:

1. Al-Hamaly MA, Cox AH, Haney MG, et al. Zebrafish drug screening identifies Erlotinib as an inhibitor of Wnt/β-catenin signaling and self-renewal in T-cell acute lymphoblastic leukemia. *Biomed Pharmacother*. 2024;170:116013. doi:10.1016/j.biopha.2023.116013

2. An LL, Zhao DF, Hou RF, et al. [Treatment response of a two-dose regimen of dose-adjusted inotuzumab ozogamicin in relapsed/refractory B-cell acute lymphoblastic leukemia]. *Zhonghua Xue Ye Xue Za Zhi Zhonghua Xueyexue Zazhi*. 2023;44(11):911-916. doi:10.3760/cma.j.issn.0253-2727.2023.11.005

3. Kato M, Manabe A. Treatment and biology of pediatric acute lymphoblastic leukemia. *Pediatr Int Off J Jpn Pediatr Soc*. 2018;60(1):4-12. doi:10.1111/ped.13457

4. Antić Ž, Lentes J, Bergmann AK. Cytogenetics and genomics in pediatric acute lymphoblastic leukaemia. *Best Pract Res Clin Haematol*. 2023;36(4):101511. doi:10.1016/j.beha.2023.101511

5. Barrios-Palacios D, Organista-Nava J, Balandrán JC, et al. The Role of miRNAs in Childhood Acute Lymphoblastic Leukemia Relapse and the Associated Molecular Mechanisms. *Int J Mol Sci*. 2023;25(1):119. doi:10.3390/ijms25010119

6. Borowitz MJ, Wood BL, Devidas M, et al. Prognostic significance of minimal residual disease in high risk B-ALL: a report from Children's Oncology Group study AALL0232. *Blood*. 2015;126(8):964-971. doi:10.1182/blood-2015-03-633685

7. Cabanillas ME, Kantarjian H, Thomas DA, et al. Epoetin alpha decreases the number of erythrocyte transfusions in patients with acute lymphoblastic leukemia, lymphoblastic lymphoma, and Burkitt leukemia/lymphoma: Results of a randomized clinical trial. *Cancer*. 2012;118(3):848-855. doi:10.1002/cncr.26341

8. Conter V, Bartram CR, Valsecchi MG, et al. Molecular response to treatment redefines all prognostic factors in children and adolescents with B-cell precursor acute lymphoblastic leukemia: results in 3184 patients of the AIEOP-BFM ALL 2000 study. *Blood*. 2010;115(16):3206-3214. doi:10.1182/blood-2009-10-248146

9. Feldman K, Aaronson K, Gu T, et al. Desensitization using pegaspargase in the era of commercially available *Erwinia* : A single-institution report on efficacy, cost, and resource utilization. *Pediatr Blood Cancer*. 2024;71(4):e30891. doi:10.1002/pbc.30891

10. Foà R, Bassan R, Elia L, et al. Long-Term Results of the Dasatinib-Blinatumomab Protocol for Adult Philadelphia-Positive ALL. *J Clin Oncol*. 2024;42(8):881-885. doi:10.1200/JCO.23.01075

11. Gooptu M, Kim HT, Howard A, et al. Effect of Sirolimus on Immune Reconstitution Following Myeloablative Allogeneic Stem Cell Transplantation: An Ancillary Analysis of a Randomized Controlled Trial Comparing Tacrolimus/Sirolimus and Tacrolimus/Methotrexate (Blood and Marrow Transplant Clinical Trials Network/BMT CTN 0402). *Biol Blood Marrow Transplant*. 2019;25(11):2143-2151. doi:10.1016/j.bbmt.2019.06.029

12. Bhatia S, Hageman L, Chen Y, et al. Effect of a Daily Text Messaging and Directly Supervised Therapy Intervention on Oral Mercaptopurine Adherence in Children With Acute Lymphoblastic Leukemia: A Randomized Clinical Trial. *JAMA Netw Open*. 2020;3(8):e2014205. doi:10.1001/jamanetworkopen.2020.14205

13. Conklin HM, Helton S, Ashford J, et al. Predicting Methylphenidate Response in Long-Term Survivors of Childhood Cancer: A Randomized, Double-Blind, Placebo-Controlled, Crossover Trial. *J Pediatr Psychol*. 2010;35(2):144-155. doi:10.1093/jpepsy/jsp044

14. Kauhanen L, Järvelä L, Lähteenmäki PM, et al. Active video games to promote physical activity in children with cancer: a randomized clinical trial with follow-up. *BMC Pediatr*. 2014;14:94. doi:10.1186/1471-2431-14-94

15. Lancet JE, Duong VH, Winton EF, et al. A Phase I Clinical-Pharmacodynamic Study of the Farnesyltransferase Inhibitor Tipifarnib in Combination with the Proteasome Inhibitor Bortezomib in Advanced Acute Leukemias. *Clin Cancer Res*. 2011;17(5):1140-1146. doi:10.1158/1078-0432.CCR-10-1878

16. McDonald GB, Freston JW, Boyer JL, DeLeve LD. Liver Complications Following Treatment of Hematologic Malignancy With Anti-CD22-Calicheamicin (Inotuzumab Ozogamicin). *Hepatology*. 2019;69(2):831-844. doi:10.1002/hep.30222

17. Maury S, Chevret S, Thomas X, et al. Rituximab in B-Lineage Adult Acute Lymphoblastic Leukemia. *N Engl J Med*. 2016;375(11):1044-1053. doi:10.1056/NEJMoa1605085

18. Gharial J, Guilcher G, Truong T, et al. Busulfan with 400 centigray of total body irradiation and higher dose fludarabine: An alternative regimen for hematopoietic stem cell transplantation in pediatric acute lymphoblastic leukemia. *Pediatr Blood Cancer*. 2024;71(4):e30844. doi:10.1002/pbc.30844

19. Goto H. [CAR-T therapy for adult B-cell acute lymphoblastic leukemia]. *Rinsho Ketsueki*. 2023;64(11):1432-1439. doi:10.11406/rinketsu.64.1432

20. Itoh M, Sasaki S, Tohda S. Renin-Angiotensin System Inhibitors Suppress the Growth of Leukemia Cells. *Anticancer Res*. 2024;44(3):1183-1192. doi:10.21873/anticanres.16913

21. Katoueezadeh M, Maleki P, Torabizadeh SA, et al. Combinatorial targeting of telomerase and DNA-PK induces synergistic apoptotic effects against Pre-B acute lymphoblastic leukemia cells. *Mol Biol Rep*. 2024;51(1):163. doi:10.1007/s11033-023-09087-9

22. Kumada H, Itoh M, Tohda S. Effect of Ferroptosis Inducers and Inhibitors on Cell Proliferation in Acute Leukemia. *Anticancer Res*. 2024;44(3):1003-1010. doi:10.21873/anticanres.16895

23. Li TD, Hu SY, Zhai Z, et al. [Clinical Analysis of Philadelphia Chromosome-Like Acute Lymphoblastic Leukemia in Children]. *Zhongguo Shi Yan Xue Ye Xue Za Zhi*. 2024;32(1):78-84. doi:10.19746/j.cnki.issn.1009-2137.2024.01.013

24. Lin M, Zhao X, Chang Y, Zhao X. Current assessment and management of measurable residual disease in patients with acute lymphoblastic leukemia in the setting of CAR-T-cell therapy. *Chin Med J (Engl)*. 2024;137(2):140-151. doi:10.1097/CM9.0000000000002945

25. Mizutani Y, Kusakabe S, Fukushima K, et al. [Sequential therapy with inotuzumab ozogamicin followed by CAR T-cell therapy for Philadelphia chromosome-negative acute lymphoblastic leukemia]. *Rinsho Ketsueki*. 2024;65(2):78-83. doi:10.11406/rinketsu.65.78

26. Pagliaro L, Cerretani E, Vento F, et al. CAD204520 Targets NOTCH1 PEST Domain Mutations in Lymphoproliferative Disorders. *Int J Mol Sci*. 2024;25(2):766. doi:10.3390/ijms25020766

27. Saffari N, Rahgozar S, Faraji E, Sahin F. Plasma-derived exosomal miR-326, a prognostic biomarker and novel candidate for treatment of drug resistant pediatric acute lymphoblastic leukemia. *Sci Rep*. 2024;14(1):691. doi:10.1038/s41598-023-50628-w

28. Satoh T, Kayano H, Kohri M, et al. Coincidence of *de novo* T-lymphoblastic lymphoma and cutaneous gamma/delta peripheral T-cell lymphoma. *J Clin Exp Hematop*. 2023;63(4):257-261. doi:10.3960/jslrt.23042

29. Shyr D, Davis KL, Bertaina A. Stem cell transplantation for ALL: you've always got a donor, why not always use it? *Hematology*. 2023;2023(1):84-90. doi:10.1182/hematology.2023000423

30. Wang F, Li Y, Yang Z, et al. Targeting IL-17A enhances imatinib efficacy in Philadelphia chromosome-positive B-cell acute lymphoblastic leukemia. *Nat Commun*. 2024;15(1):203. doi:10.1038/s41467-023-44270-3

31. Wu JR, Shih PC, Li C, et al. Lineage switch of KMT2A-rearranged adult B-lineage acute lymphoblastic leukemia following bispecific T-cell engager and monoclonal antibody therapy. *J Hematop*. 2023;16(2):103-109. doi:10.1007/s12308-023-00539-6

32. Yoon JH, Lee S. Diagnostic and therapeutic advances in adults with acute lymphoblastic leukemia in the era of gene analysis and targeted immunotherapy. *Korean J Intern Med*. 2024;39(1):34-56. doi:10.3904/kjim.2023.407